

Assignment 2 COMP 551

Leen Assaf, Karl Wehbe, and Vincent Barsalou

Abstract. This report presents the implementation and evaluation of **linear regression**, **logistic regression**, and **multi-class classification** models. The primary objective is to gain hands-on experience in developing these algorithms and assessing their performance on tabular datasets.

For **binary classification**, we used the **Breast Cancer** dataset containing numerical features to predict binary target variables. For **multi-class classification**, we utilized the **Palmer Penguins** dataset to predict categorical class labels.

The project involved data preprocessing, feature importance analysis, and the implementation of **Multiple linear regression** as a baseline model. **Logistic regression** and **Multi-class classification** models were developed with gradient checking and cross-entropy loss monitoring to ensure correctness.

Performance was evaluated using AUROC for binary classification and accuracy for multi-class classification. Feature ranking, ROC curves, and heatmaps provided insights into model behavior and feature importance.

In this project, we investigated the performance of **linear classification models** on two benchmark datasets and found that **logistic** and **multi-class regression** achieved higher accuracy than **simple** and **multiple linear regression** but were significantly slower to train. However, the differences in accuracy between the models were not substantial. Overall, all models demonstrated high performance, reinforcing the importance of selecting the appropriate model based on dataset characteristics and computational efficiency.

1 Introduction

Classification is a key problem in machine learning with many real-world applications, such as diagnosing diseases and categorizing objects. In this project, we explored different ways to approach classification by implementing and comparing linear regression, logistic regression, and multi-class classification models. The main goal was to understand how these models work, see how well they perform on real datasets, and analyze which features play the biggest role in making predictions.

In this project, we experimented two datasets: the Breast Cancer Wisconsin dataset for binary classification and the Palmer Penguins dataset for multi-class classification. Both datasets have structured numerical features that made it easier to compare how different models behave.

A key focus of this project was to build these models from scratch, ensuring an in-depth understanding of their inner workings. The models were trained using gradient descent, with additional validation through gradient checking and cross-entropy loss monitoring. Performance was assessed using AUROC for binary classification and accuracy for multi-class classification. Furthermore, feature importance was visualized using regression coefficients and heatmaps to better understand which features had the most impact on predictions.

2 Datasets

The datasets we used focus on two different classification problems: medical diagnosis and species classification. **The Breast Cancer Wisconsin** dataset is used in medical research to classify tumors as malignant or benign based on 30 numerical features that describe cell

characteristics from biopsy samples. This dataset is useful for evaluating classification models in a high-dimensional setting where feature selection plays an important role. **The Palmer Penguins** dataset is commonly used in ecological studies to classify penguins into three species: Adelie, Chinstrap, and Gentoo, based on four numerical features related to their physical attributes. This dataset provides a straightforward multi-class classification problem with clear class distinctions. To prepare the data, we first standardized all numerical features to ensure consistent scaling. For the Breast Cancer dataset, we removed low-impact features, which we identified through a simple linear regression analysis. In the Penguins dataset, we dropped categorical features like **island** and **sex**, as they were not relevant to the classification. Both datasets are relatively balanced, which helps ensure that model performance is not biased toward any particular class. The Breast Cancer dataset presents a higher-dimensional challenge, making it a good test case for feature selection, while the Penguins dataset offers a simpler but well-separated classification problem that allows for intuitive model interpretation. By working with both datasets, we were able to analyze how different models handle binary and multi-class classification and how feature selection impacts predictive performance.

3 Results

3.1 Feature Importance

In this section, we rank and visualize the features based on their coefficients obtained from a **simple linear regression** model. For the **Breast Cancer** dataset, we applied a simple linear regression model. The features represent various measurements derived from digitized images of fine needle aspirate (FNA) of breast masses

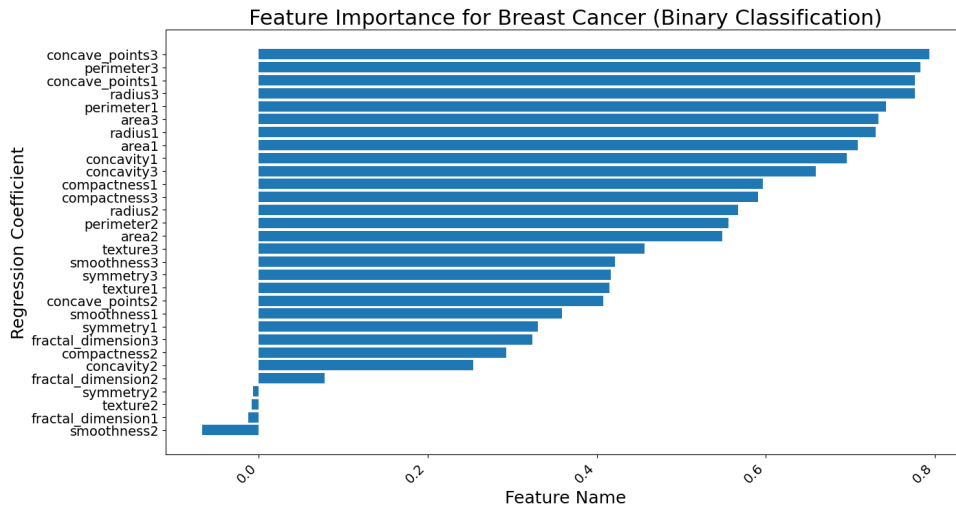


Figure 1: Feature importance for Breast Cancer (Binary Classification) dataset from the Simple Linear Regression model.

Figure 1 presents a horizontal bar plot of the regression coefficients, with the feature names on the y-axis and their corresponding coefficient weights on the x-axis. The bars are sorted in descending order, placing the most positively weighted features at the top. Higher positive coefficients suggest a stronger positive association with the cancerous class (i.e., larger values of that characteristic are more indicative of a tumor/positive label). The negative coefficients suggest an inverse relationship and are then removed from the dataset.

For the **multiclass classification** task on the **Penguin dataset**, we trained a simple linear regression model with one-hot encoded target variables for the three classes (Adelie, Chinstrap,

Gentoo). Each class has its own set of regression coefficients, capturing how each feature influences the likelihood of belonging to that class versus the others.

Below are three horizontal bar plots (Figures 2 – 4), each corresponding to one species. The x-axis shows the regression coefficient weight, and the y-axis lists the features. Positive coefficients indicate that higher values of that feature increase the likelihood of that particular class. Negative coefficients suggest that higher values of that feature decrease the likelihood of that class (i.e., the feature value is more indicative of another class). In this case, a given feature may be important for one species and less so for another, meaning that each feature provides unique discriminative information. Consequently, no single feature can be dropped without risking a loss of valuable class-specific insights.

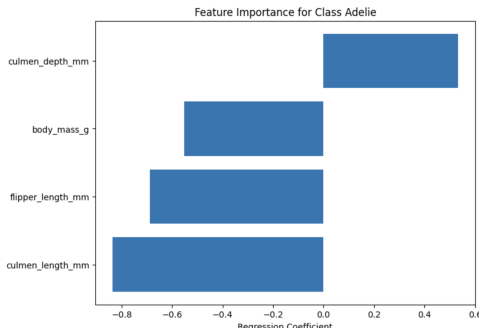


Figure 2: Feature importance for Adelie

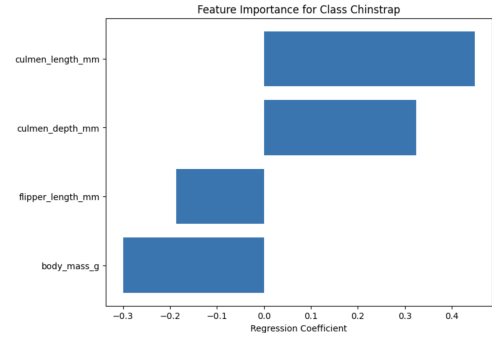


Figure 3: Feature importance for Chinstrap

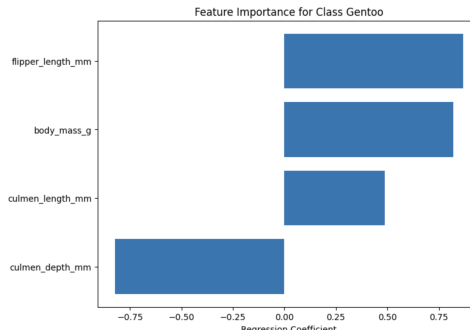


Figure 4: Feature importance for Gentoo

3.2 Gradients for Logistic Regression

In this part, we verify the correctness of our **binary logistic regression** and **multi-class logistic regression** model implementations through gradient checking and by monitoring the training cross-entropy loss. For both models, we validated the analytical gradient computations by comparing them with approximate gradients obtained using a small numerical perturbation approach. The numerical difference between the analytical and approximate gradients was found to be approximately 5.5×10^{-11} for the **binary classification** task and 9.6×10^{-9} for the **multi-class classification** task — both well below the acceptable threshold of 10^{-6} . These extremely small differences confirm that our gradient computations are correct, and the gradient check passed successfully.

3.3 Convergence of Single and MultiNomial Logistic Regression

This section presents the results from training **logistic regression** on the **Breast Cancer** dataset (binary classification) and **multinomial logistic regression** on the **Penguin** dataset (multiclass classification). Convergence plots were used to assess how effectively the models minimized cross-entropy loss over 1000 iterations with a learning rate of $\eta = 0.01$, which was selected to balance learning speed and stability.

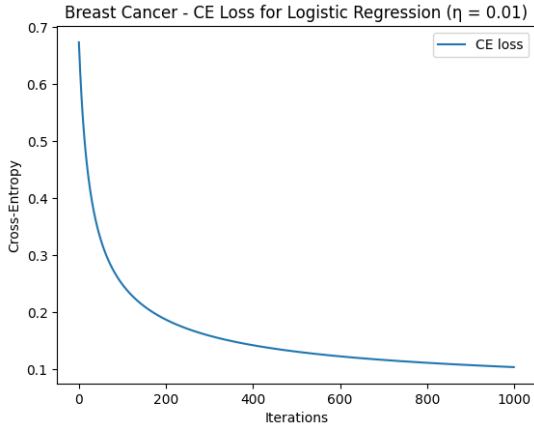


Figure 5: Breast Cancer Convergence Plot

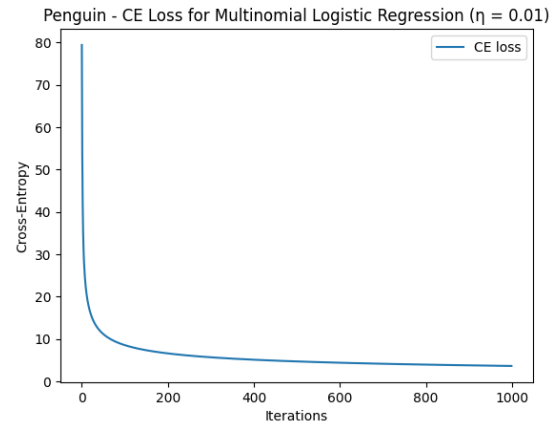


Figure 6: Penguin Convergence Plot

In both cases, the models showed smooth and stable convergence, with rapid initial reductions in loss followed by slower improvements as they approached a minimum. This behavior is characteristic of well-tuned learning rates, where the models can efficiently learn without overshooting or diverging. The **logistic regression** model on the **Breast Cancer** dataset achieved a lower final loss compared to the **multinomial logistic regression** model on the **Penguin** dataset, which is expected due to the increased complexity of distinguishing between more categories. Overall, these results demonstrate that the chosen learning rate was reasonably effective, allowing both models to converge steadily without instability.

3.4 Binary Classification dataset - ROC comparison of Logistic and Linear regression

In this part, we compare the performance of the **linear regression** models to the previously trained **logistic regression** models using the same test set. The comparison is based on two key evaluation metrics: **AUROC** and **prediction accuracy**, which together provide insight into both overall correctness and the models' ability to distinguish between classes.

For the **binary classification** task on the **Breast Cancer** dataset, we evaluated three models: **Simple Linear Regression (SLR)**, **Multiple Linear Regression (MLR)**, and **Binary Logistic Regression (BLR)**. The test accuracy and AUROC for each model are summarized below:

- **Simple Linear Regression (SLR):** Accuracy = **92.40%**, AUROC = **0.92**
- **Multiple Linear Regression (MLR):** Accuracy = **97.08%**, AUROC = **0.97**
- **Binary Logistic Regression (BLR):** Accuracy = **97.66%**, AUROC = **0.97**

We further compared these models by plotting their ROC curves on the same figure. As shown in Figure 7 below, both MLR and BLR achieve the highest AUROC of 0.97, while SLR falls

behind with an AUROC of 0.92. These results indicate that both MLR and BLR provide strong classification performance for this dataset and can be considered effective options in practice.

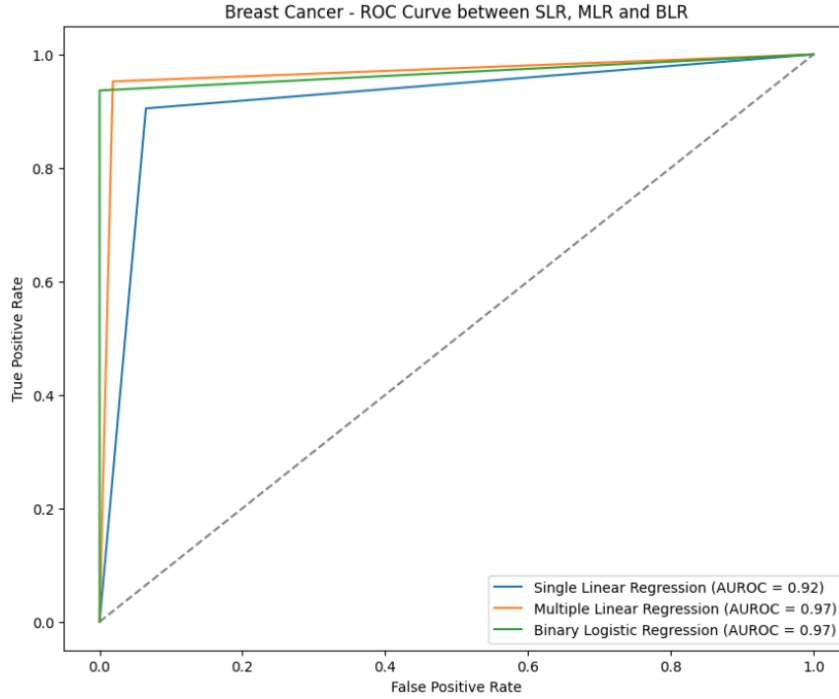


Figure 7: Breast Cancer - ROC Curve between Single and Multiple Linear Regression, and Binary Logistic Regression.

3.5 Multiclass dataset - Model accuracy

In this part, we report and compare the test accuracies of the **multivariate linear regression** (MVLRL) model and the **multiple linear regression** (MCLR) model on the multi-class classification test dataset. By evaluating both models on the same test set, we can assess their relative effectiveness in handling multi-class classification tasks. This comparison highlights how well each model generalizes to unseen data and provides insight into the suitability of linear and logistic approaches for multi-class problems.

For the multi-class classification experiments on the test data, the following outcomes were observed: the **simple linear regression** (SLR) model achieved an accuracy of **93.69%**, which is reasonable but indicates some misclassifications. In contrast, both the **multivariate multiple linear regression** (MVLRL) and the **multi-class logistic regression** (MCLR) models achieved perfect accuracy, each reaching **100.00%**. These results suggest that MVLRL and MCLR provide exceptionally strong performance on this dataset clearly outperforming SLR.

3.6 Binary Classification dataset - Feature importance for different models

In this part, we compare the regression coefficients learned by the **multiple logistic regression** (MLR) model with the coefficients obtained from the **simple linear regression** (SLR) and **multiple linear regression** (MLR) models. Differences in the top-ranked features may emerge due to the differing optimization objectives of linear regression, which minimizes **mean squared error** (MSE), versus logistic regression, which maximizes the **log-likelihood** of correct class predictions. These differences provide valuable insights into how each model interprets and prioritizes features when solving the binary classification task.

In linear regression (both SLR and MLR), the coefficients are estimated by minimizing the MSE between predicted values and the numerically encoded class labels. This objective tends

to favor features that reduce overall variance, even if they do not necessarily create the clearest separation between classes. In contrast, logistic regression directly optimizes the log-likelihood, which places greater emphasis on features that improve the separability between the two classes. As a result, the most influential features in linear regression may differ from those in logistic regression. A feature that reduces MSE effectively might not enhance class separability as much as another feature that logistic regression highlights as critical for distinguishing between classes.

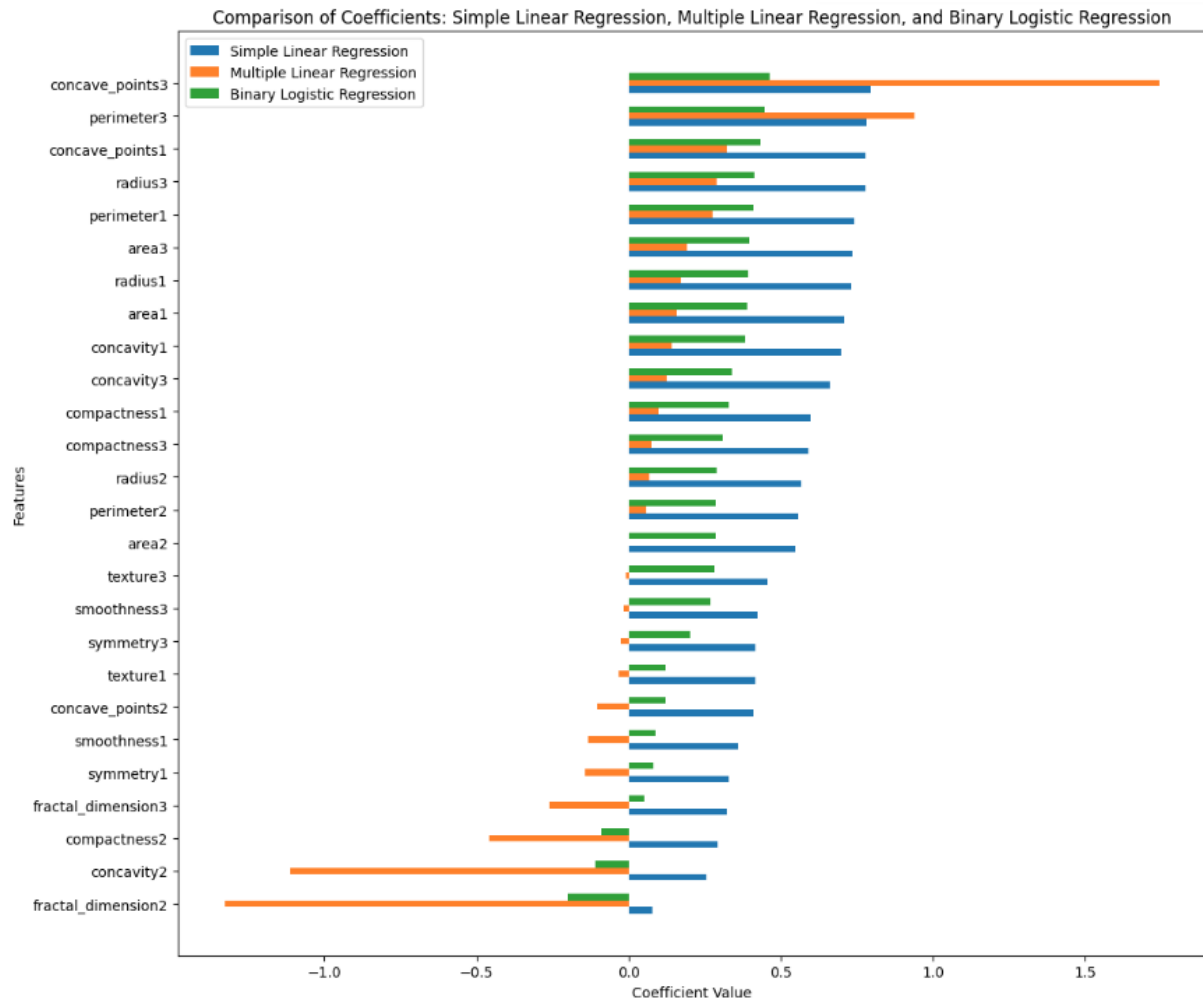


Figure 8: Comparison of Coefficients: Simple Linear Regression, Multiple Linear Regression, and Binary Logistic Regression.

In Figure 8, Simple and Multiple Linear Regression show similar trends, as both minimize error using least squares. In contrast, Binary Logistic Regression assigns different magnitudes — and sometimes opposite signs — to some features, since it models the probability of a binary outcome using the sigmoid function. The top features in both regression models (such as concave points, perimeter, radius, and area) remain important in Binary Logistic Regression, though their relative importance shifts as logistic regression optimizes for classification rather than minimizing error. Overall, some features remain key across all models, but their influence varies between regression and classification tasks.

3.7 Multiclass dataset - Comparison between weight matrices

In this section, we present a set of **heatmaps** illustrating how different features contribute to each class under various **regression models** for the **Penguin dataset**. We compare the coefficients from our **multi-class logistic regression** model with those from **simple linear regression** and **multiple linear regression** to investigate whether the most influential features remain consistent and how their relative importance might shift.

Comparing the heatmaps, the top features per class remain largely consistent; however, their intensity and dominance over other features vary. Some features become more dominant (e.g., `culmen_length_mm` for the Chinstrap class), while others become less so (e.g., `culmen_depth_mm` for the Adelie class). For instance, a feature that moderately lowers the MSE in linear regression might become highly influential in logistic regression if it strongly discriminates one species from another (e.g., `body mass` for the Adelie class). These differences arise because linear regression and logistic regression optimize different objectives. Linear regression minimizes the squared difference between predicted and actual numerical labels, whereas logistic regression maximizes the probability of correct class predictions by focusing on class boundaries. Thus, while the overall ranking of features may be similar, the exact order and intensity of their coefficients can differ, reflecting the distinct optimization goals of the two methods.

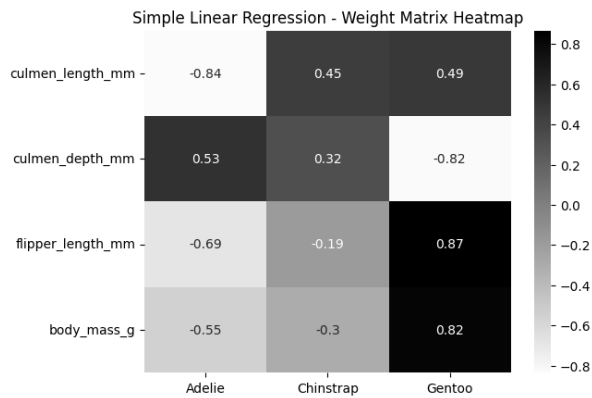


Figure 9: Single Linear Regression heatmap

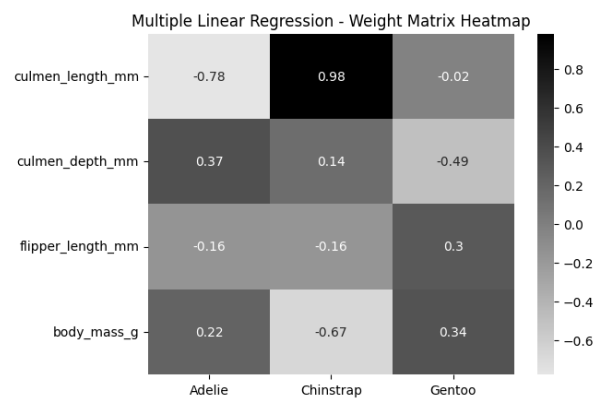


Figure 10: Multiple Linear Regression heatmap

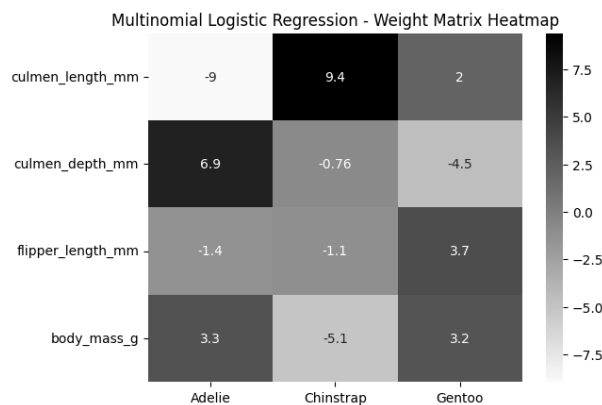


Figure 11: Multinomial Logistic Regression heatmap

4 Conclusion

Our results showed that logistic regression consistently outperformed linear regression for classification tasks. This makes sense because linear regression minimizes the mean square error (MSE), which is not designed for classification, whereas the logistic regression optimizes for class separation. In the BReast Cancer dataset, both logistic regression and multiple linear regression achieved high accuracies (97%), but simple linear regression performed worse (92%) due to its inability to properly model probabilities. In the Penguin dataset, the logistic and multivariate linear regression models both achieved perfect accuracy, showing that the dataset had clear class separability.

One key observation was that logistic regression ranked the importance of features similarly to linear regression but assigned more extreme coefficients to important features. This happens because logistic regression prioritizes features that help separate classes rather than just minimizing the variance. For example, in the Breast Cancer dataset, features related to cell texture and radius had strong predictive power, while in the Penguins From a computational standpoint, linear regression is faster to train because it has a closed-form solution, whereas logistic regression relies on iterative optimization (gradient descent), making it slower. However, the improved classification performance of logistic regression justifies the extra computation especially for datasets where proper class separation is essential.

Overall, our findings confirm that while linear regression can technically be used for classification, logistic regression is the better choice due to its probabilistic approach and better decision boundaries. The feature importance analysis also reinforced how different models interpret variables differently, providing useful insights into what makes an attribute valuable for classification.

5 Statement of contribution

Leen Assaf: Task 2, 3, 4 and Report

Karl Wehbe: Task 2, 3, 4 and Report

Vincent Barsalou: Task 1